Analytical Cumulative Exam
April 25, 1998

The expression for rate of build up of activity in a sample in the nuclear reactor during a neutron activation analysis experiment is:

$$\frac{dN}{dt} = N_t \, \sigma \, \phi \, (1 - e^{-\lambda t})$$

(a) (20) Explain each term and say how and why it influences the amount of activity induced.

(b) (20) Sketch diagrams showing how the activity varies with time in the reactor and after removal from the reactor for (i) ^{56}Mn and (ii) ^{59}Fe.

(c) (40) A meteorite contains 0.30 wt % Cu and 0.0053 wt % Co. A reactor producing a neutron flux of 5 x 10^{11} cm^{-2} sec^{-1} is available, but it takes 0.7 d to ship the samples from the reactor to the lab. Assuming 10 mg samples are irradiated, and it is possible to make separate irradiations for Cu and Co, how long should the irradiations be if 10,000 counts/minute are needed?

(d) (20) How long does one need to count the samples in part (b) in order to achieve a 1 σ of 1% ?

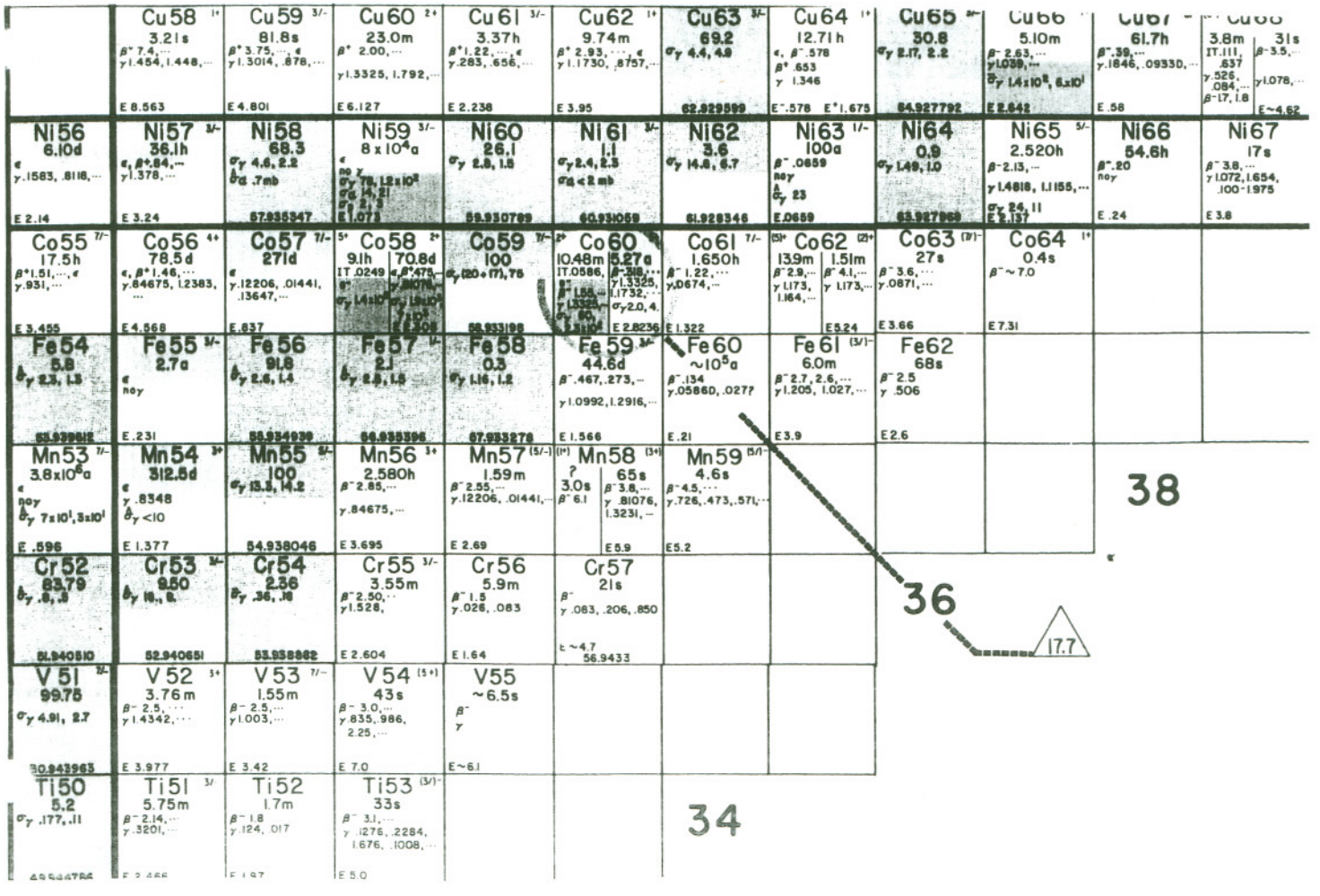

TABLE OF NUCLIDES

Z	El	A	Abundance or $t_{1/2}$	Decay Mode	Δ(MeV)	Jπ	σ_n(b)
0	n	1	10.6 m	β^-,no γ	8.071	1/2+	
1	H	1	*99.985%*		7.289	1/2+	0.332
		2	*0.0148%*		13.136	1+	5.2×10^{-4}
		3	12.33 y	β^-,no γ	14.950	1/2+	$<6\times10^{-6}$ sc
2	He	3	*1.38×10^{-4}%*		14.931	1/2+	5.33×10^{3} np
		4	*99.99986%*		2.425	0+	
		6	0.808 s	β^-,no γ	17.597	0+	
		8	0.122 s	β^-,β^-n 12%	31.609	0+	
3	Li	6	*7.5%*		14.087	1+	942 na
		7	*92.5%*		14.908	3/2−	0.045 rs
		8	0.84 s	$\beta^-2\alpha$	20.947	2+	
		9	0.178 s	β^-,β^-n2α 35%	24.955	(3/2)−	
		11	8.5 ms	β^-,β^-n 61%	40.94		
4	Be	7	53.3 d	EC	15.770	3/2−	5×10^{4} rs np
		9	*100%*		11.348	3/2−	0.008
		10	1.6×10^{6} y	β^-,no γ	12.608	0+	<0.001 rs
		11	13.8 s	β^-,$\beta^-\alpha$ 3%	20.176	1/2+	
		12	11.4 ms	β^-,β^-n	25.03	0+	
5	B	8	0.769 s	$\beta^+2\alpha$	22.922	2+	
		10	*19.8%*		12.052	3+	3838 na
		11	*80.2%*		8.668	3/2−	0.005 rs
		12	20.4 ms	β^-,$\beta^-3\alpha$ 1.6%	13.370	1+	
		13	17.4 ms	β^-,β^-n 0.28%	16.562	3/2−	
		14	16 ms	β^-	23.657	2−	
6	C	9	0.1265 s	β^+p2α	28.912	(3/2−)	
		10	19.2 s	β^+	15.703	0+	
		11	20.38 m	β^+ 99.76%, EC 0.24%,no γ	10.650	3/2−	
		12	*98.89%*		0	0+	0.0034
		13	*1.11%*		3.125	1/2−	9×10^{-4}
		14	5730 y	β^-,no γ	3.020	0+	$<1\times10^{-6}$ rs
		15	2.449 s	β^-	9.873	1/2+	
		16	0.75 s	β^-n >98.8%	13.693	0+	
7	N	12	11.0 ms	β^+,$\beta^+3\alpha$ 3.5%	17.338	1+	
		13	9.96 m	β^+,no γ	5.346	1/2−	
		14	*99.63%*		2.863	1+	1.82 sc np
		15	*0.366%*		0.102	1/2−	4×10^{-5} rs
		16	7.13 s	β^-,$\beta^-\alpha$ 0.0012%	5.682	2−	
		17	4.17 s	β^-,β^-n 95%	7.870	1/2−	
		18	0.63 s	β^-	13.274	0,1,2−	
8	O	13	8.9 ms	β^+p	23.105	3/2−	
		14	70.60 s	β^+	8.008	0+	
		15	122 s	β^+ 99.89%, EC 0.11%,no γ	2.855	1/2−	
		16	*99.76%*		−4.737	0+	1.8×10^{-4} rs
		17	*0.038%*		−0.810	5/2+	0.235 sc na
		18	*0.204%*		−0.783	0+	1.6×10^{-4}
		19	26.9 s	β^-	3.331	5/2+	
		20	13.5 s	β^-	3.799	0+	
9	F	17	64.5 s	β^+,no γ	1.952	5/2+	
		18	109.8 m	β^+ 96.9%, EC 3.1%,no γ	0.872	1+	
		19	*100%*		−1.487	1/2+	0.010 rs
		20	11.0 s	β^-	−0.017	2+	
		21	4.32 s	β^-	−0.047	5/2+	
		22	4.23 s	β^-	2.826	4+	
		23	2.2 s	β^-	3.35	(5/2)+	
10	Ne	17	0.109 s	β^+p	16.478	1/2−	
		18	1.67 s	β^+	5.319	0+	
		19	17.3 s	β^+ 99+%,EC 0.102%	1.751	1/2+	
		20	*90.51%*		−7.043	0+	0.038 rs
		21	*0.27%*		−5.733	3/2+	0.7 rs